



Foundational questions on artificial and biological intelligences

We define intelligent systems as those that achieve their goals successfully, even in situations that they haven't encountered before. At Lossfunk, we want to study how intelligence manifests in biological systems and then use that inspiration to build artificial systems that go beyond where current AIs are.

[Research direction 1] AI that adapts to a domain

We're interested in exploring how to build AI systems that learn on-the-fly whatever is specific to a domain and start outperforming relevant domain experts. Our bet is that a narrow AI that adapts with the user will eventually replace the current breed of "general" AI/LLMs that are fixed for everyone. This is because the world is full of locally-relevant details and nuances which an AI system should be able to learn. This learning requires recognizing domain-specific learning signals from mere noise. Our current work has established that LLMs perform badly in zero-shot manner for out-of-distributions such as esoteric languages, but if you put them in agentic loops, they experiment, take notes and eventually find a way to perform. We're excited to explore and create such AIs that adapt on the fly to all relevant out-of-domain problems that are thrown at it.

Topics: continual learning, memory, test time adaptation, active learning, sample efficiency, efficient training or inference, personalization, curiosity, exploration, agency, autonomy, OOD generalization, curriculum learning, meta-learning, uncertainty modeling

Some example questions:

- What does it mean to "understand" a domain, and how does that differ from pattern matching over training data?
- What kind of memory should an adapting AI have? What should be baked in weights or assembled during inference (via files or context)?
- What techniques could enable minimal catastrophic forgetting as the AI learns something new in a domain?
- What's the right way to model a domain? What should the world model look like? What should be parametric or non-parametric?
- How can training/learning happen locally in a constrained compute environment?

[Research direction 2] Creativity in artificial systems

We're interested in why AI systems produce average outputs despite having ingested extraordinary creative work. Our bet is that creativity requires structured representations of possibility spaces; not just exposure to examples, but understanding of the domain's

structure well enough to identify where unexplored territory lies. For instance, a creative artist doesn't just know prior art. They understand the constraints and possibilities of their medium + what has been done before well enough to find setups nobody has exploited yet. We're investigating what computational objects enable this. Our current work revolves around investigating research taste in LLMs and previously we investigated jokes production ability of LLMs. We're not satisfied with where things stand, and want to build the next generation of AI systems that expand a domain (instead of operating within the confines of their training).

Topics: novelty, creativity, representations, data manifold, extrapolation, surprise, world models, recombination, concept modeling, scientific theory building, innovation, abstractions, program synthesis, knowledge representation, taste

Some example questions:

- How should novelty be modeled, detected and measured? What differentiates it from mere noise or surprising but irrelevant detail?
- What role do world models and imagination play in creativity?
- What process do most creative people in different domains follow and how can we encode that into AI?
- What is "good taste" in a domain? What contribution does mere popularity/luck have in it v/s genuinely better process/output?

[Research direction 3] Biological foundations of intelligence

We're interested in how biological systems — from single organisms to evolutionary processes — exhibit intelligence by detecting, prioritising, and responding to novelty. Our bet is that the mechanisms biology has evolved for handling the unknown (including consciousness, language and culture) reveal general principles about intelligence that current AI architectures are missing. For example, we believe consciousness, rather than being a mysterious byproduct, plays a functional role specifically in situations where an organism's predictive models fail and novel action selection is required. Our early work states our philosophical orientation toward pragmatism.

Topics: brain, evolution, consciousness, language, culture, qualia, representations, cooperation, efficiency, artificial life, origin of life, developmental biology, entropy, attractors, rationality, game theory, neuroscience, cognitive science

consciousness, predictive processing, evolution, culture, artificial life, language (origins), developmental cognition, neuroscience, representations

Some example questions:

- What computational signatures distinguish conscious novelty-processing from unconscious routine?
- Are there universal patterns in how evolving systems respond to environmental novelty? Does something with similar functional signatures as our qualia space or language necessarily emerge during evolution of artificial agents?

- How does evolution and human culture accumulate what works? How do we create that in artificial systems?
- How do biological brains represent concepts for flexible recombination?